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SUPERSEDING
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MILITARY SPECIFICATION

CHROMIC CHLORIDE, HEXAHYDRATE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers reagent grade chromic chloride, hexahydrate.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.

MILITARY

MIL-P-15011	-	Pallets, Material Handling, Hardwood, Post Construction, 4-Way.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Appearance. Chromic chloride, hexahydrate shall be dark green granules or lumps.

3.2 Chemical characteristics. Chromic chloride, hexahydrate shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as Cr)	18.9	-----	4.2.4.1
Insoluble matter	----	0.010	4.2.4.2
Sulfate (SO_4)	----	0.020	4.2.4.3
Iron (Fe)	----	0.010	4.2.4.4
Substances not precipitated by ammonium hydroxide	----	0.30	4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Regular production.

4.2.1 Lotting. A lot shall consist of the chromic chloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling for test. Sampling shall be conducted in accordance with MIL-STD-105, level S-2. Approximately equal portions of chromic chloride shall be taken from each container in the sample, composited, and placed in a suitable clean, dry container labeled to identify the lot and the containers from which it was taken. The total composited sample shall be no less than 1/4 pound.

4.2.3 Inspection. The composited sample of chromic chloride shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as Cr). Weigh to the nearest 0.1 milligram (mg) approximately 1.5 grams (g) of the specimen. Dissolve in water and dilute in a volumetric flask to 100 milliliters (ml). Transfer 25 ml of the solution to a 300-ml flask, heat to 80° ± 2°C, and cautiously add, in small portions, 2 g of sodium peroxide. Dilute with 100 ml of water. Boil gently until the volume is reduced to approximately 50 ml or until the excess of peroxide is destroyed. Dilute to approximately 150 ml with water, add 7 ml of 85-percent phosphoric acid, and cool. Add 3 g of potassium iodide, allow to stand for 30 minutes, then titrate the liberated iodine with 0.1N sodium thiosulfate solution, adding starch toward the end as indicator. Calculate the percent chromium as follows:

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$$\text{Percent chromium} = \frac{1.734AB}{W}$$

where: A = Normality of sodium thiosulfate solution,
B = Milliliters of sodium thiosulfate solution used, and
W = Weight of specimen in grams.

4.2.4.2 Insoluble matter. Weigh to the nearest 0.1 mg approximately 10 g of the specimen. Dissolve in 100 ml of water and heat on a steam bath for 30 minutes. Filter through a tared filter crucible, wash with hot water until the washings are colorless, and dry the residue at 105° ± 2°C for one hour. Cool, and weigh. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100(A-B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.3 Sulfate.

(a) Potassium sulfate solution. Dissolve 0.1814 g of potassium sulfate in water to make 1 liter of solution.

(b) Procedure. Dissolve 3.000 g of the specimen in 130 ml of water, heat to boiling, and add sufficient 28-percent ammonium hydroxide solution to precipitate all the chromium. Boil to expel the excess of ammonia, filter, amd wash with hot water to make 150 ml of filtrate. Eveporate 50 ml of the filtrate to 15 ml and neutralize with 10-percent hydrochloric acid. Save an additional 50 ml of the filtrate for 4.2.4.5. Add 1 ml of 1N hydrochloric acid, dilute to 25 ml, and add 2 ml of 12-percent barium chloride solution. Prepare a control with 2 ml of potassium sulfate solution prepared as specified in (a), 1 ml of 1N hydrochloric acid and 2 ml of 12-percent barium chloride solution, in the same final volume as the specimen. The turbidity produced in the specimen test shall be no greater than that in the control.

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4.2.4.4 Iron.

(a) Butanolic potassium thiocyanate solution. Dissolve 10 g of potassium thiocyanate in 10 ml of water. Warm the solution to 25° ± 2°C and add n-butanol to make 100 ml. Shake vigorously until the solution is clear.

(b) Ferric ammonium sulfate solution. Dissolve 0.1727 g of ferric ammonium sulfate in 10 ml of 10-percent sulfuric acid and dilute with water to make 2000 ml. One ml of this solution equals 0.01 mg of iron.

(c) Procedure. Dissolve 1.000 g of the specimen in 15 ml of water, add 2 ml of hydrochloric acid, and boil for two minutes. Cool, and dilute to 50 ml. To 10 ml of the solution, add 1 ml of hydrochloric acid and approximately 30 mg of ammonium persulfate and follow with 15 ml of butanolic potassium thiocyanate solution prepared as specified in (a). Shake well for 30 seconds and allow to separate. Any red color produced in the clear butanol (upper) layer shall be no darker than that produced when 2 ml of ferric ammonium sulfate solution prepared as specified in (b) is substituted for the 10 ml of specimen solution and tested in the same manner.

4.2.4.5 Substances not precipitated by ammonium hydroxide. Evaporate 50 ml of the filtrate from the test for sulfate (4.2.4.3) with 5 drops of 96-percent sulfuric acid and ignite to constant weight of residue. Calculate the percent substances not precipitated by ammonium hydroxide as follows:

$$\text{Percent substances not precipitated by ammonium hydroxide} = \frac{300A}{B}$$

where: A = Weight of residue in grams and
B = Weight of specimen in 4.2.4.3 in grams.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Chromic chloride shall be packaged level A or C as specified (see 6.2).

5.1.1 Level A. One-quarter pound ± 0.05 ounce of chromic chloride shall be packaged in either a type I, class 1 bottle utilizing inclosure b (bottles in metal cans) in accordance with the requirements for level A packaging as specified in PPP-C-301 or in a class 2, style 2 plastic bottle utilizing closure A or B and outer seal A band or tape of PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of chromic chloride, packaged as specified in 5.1.1, shall be intermediately packaged upright in a three bottle by four bottle pattern in a snug-fitting box conforming to style RSC, grade W5c of PPP-B-636. The box shall be closed in accordance with the closure requirements for class weather-resistant boxes by taping as specified in the appendix to PPP-B-636.

5.1.2 Level C. Unless otherwise specified in the contract or order, a 1/4-pound quantity of chromic chloride shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use (see 6.2).

5.2 Packing. Chromic chloride shall be packed level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four boxes of chromic chloride, intermediately packaged as specified in 5.1.1.1, shall be packed for shipment in a nailed wood box conforming to class 2, style 4-1/2, for a type 2 average load of PPP-B-621. The box shall be closed and strapped using galvanized strapping in accordance with the applicable requirements in the appendix to PPP-B-621.

5.2.2 Level B. Chromic chloride shall be packed as specified in 5.2.1, except that the box shall conform to either style RSC, grade V3c, for a type 2 average load of PPP-B-636 or style optional, class 2, for a type 2 average load of PPP-B-585 as specified (see 6.2). Closure shall be in accordance with the requirements for closure in the appendix to the applicable box specification.

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5.2.3 Level C. Chromic chloride, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to the special marking for unit containers specified in 5.3.1 and any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following special information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis or maximum limits of impurities.
(f) Precautionary marking: HARMFUL IF SWALLOWED.

5.4 Palletization. When specified in the contract or order, shipping containers of 5.2 shall be palletized in accordance with the applicable requirements of MIL-STD-147 utilizing the hardwood, 4-way, 40 by 48 inch, double-wing pallet of MIL-P-15011 (see 6.2).

6. NOTES

6.1 Intended use. Chromic chloride is intended for use as a general laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Level C unit quantity other than 1/4 pound, if required.
(d) Type of level B packing box, if required.
(e) Special marking other than 5.3.1, if required.
(f) Palletization, if required.

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6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Air Force - 68

Review activities:

Air Force - 68
DSA - GS

User activities:

Army - SM

Preparing activity:

Army - MU (EA)

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